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Memory and Attention Support Mobile Applications: A Comprehensive Review and Proposed Design Framework

ABSTRACT

This paper examines smartphone applications designed for individuals who experiences challenges related to memory and attention, including conditions such as mild cognitive impairment, dementia, Attention Deficit Hyperactivity Disorder (ADHD), as well as traumatic brain injury sequelae. This paper focuses on four key domains. Namely, effectiveness of cognitive rehabilitation applications, digital interventions targeting ADHD, outcomes of computerized memory training and Human Computer Interaction (HCI) design bases for cognitively impaired users. Based on the evidence reviewed, applications that were considerately designed has proven to enhance working memory capacity as well as attention sustainability (Cha, 2024; Chan et al., 2024; Li et al., 2026). Hence, the Memory and Attention Support Application (MASA) is proposed. It incorporates customizable reminders, an adaptive focus timer, cognitive training exercises and a minimalist and accessibility interface.

Keywords: **memory support application; attention support; cognitive rehabilitation; ADHD; human-computer interaction; assistive technology**

1. INTRODUCTION

Cognitive health has increasingly become more challenging in our daily life. Conditions spanning from pediatric ADHD to age-related neurodegeneration has affected populations in all corners of the world. However, modern digital environments cause additional stress by persistently delivering notifications and overloading us with excessive information. Smartphones present itself contradicting. On one hand, it fragments our focus with the constant switching between messages, applications, videos etc., which in return leads to a harder ability to center our attention. On the other hand, smartphones are a tool to provide support as smartphones can deliver mental health or cognitive training support. In other words, smartphones grant users with a feeling of familiarity, especially when going to a clinic or hospital. Despite a market flooded with brain-training applications, a gap between applications that are commercially available and applications genuinely backed by evidence remains. Hence, the primary concern of this paper is to bridge said gap.

2. PROBLEM STATEMENT

The demand for evidence-backed cognitive instruments has exceeded the current provision of the market. To state it another way, many applications aimed to boost productivity were humbly not designed to accommodate users with impaired cognition. These applications pile on complexity and visual clutter, which produces an interface overhead that strains a budget of limited attention (Skulmowski & Xu, 2021). Although well-thought-of clinical rehabilitation programs exist, they infrequently get implemented routinely in our lives. Consistent research has discovered that personalization is a key ingredient for cognitive enhancement but then again applications on the market largely ignore this by defaulting to rigid and general approaches that regularly dishearten the users whom they are designed to serve (Cha, 2024; Liu et al., 2024).

3. LITERATURE REVIEW

3.1 Cognitive Foundations

Working memory exists as a mental space to process active information. Once it falters, tasks performed routinely such as understanding a paragraph, planning a schedule, deciding on

what to have for lunch can transform into struggles. The occurrence of such event closely interferes with our ability to sustain attention as well as impair executive function. These are known as the characteristics of ADHD and other cognitive conditions. Hence, it should be understood that how easy an interface is for the brain to process is parallel to the visual aesthetics, especially for interface designers. Research on digital learning environments confirms that additional complexity of an interface has a direct connection to suppressed performance, playing the role of an obstacle bridge (Skulmowski & Xu, 2021).

3.2 Mobile Cognitive Rehabilitation

Shown by evidence, mobile-based cognitive rehabilitation has considerably gain support in recent years. A review conducted by Cha (2024) has found that most studies report favorable results from cognitive rehabilitation applications. The result of a combination of studies further shows that, among individuals with mild cognitive impairment, Computerized Cognitive Training (CCT) can improve several types of memory such as verbal, visual and working memory (Chan et al., 2024). Furthermore, researches have discovered that several key factors of these improvements include the number of training instances as well as the duration of those training instances, highlighting the vitality of curating and adjusting schedules to each person (Liu et al., 2024). Particularly in older adults, participation in long-term cognitive training has provided persuasive results for faster processing information speed and stronger ability to remember personal events, although existing follow-up data remains scarce (Maggio et al., 2023).

3.3 Digital Interventions for ADHD

Digital interventions for ADHD are more challenged. Early reviews have suggested valid skepticism towards marketed applications, many of which is due to the weak scientific foundations present (Gabarron et al., 2025). However, more findings of late are considerably more encouraging, especially when combining digital instruments with behavioral support as compared to unrelated or solitary substitutions. To elaborate, a combined approach such as software

integrated with proper human guidance has been shown to outperform either strategy if it were standalone.

3.4 HCI Design and AI Personalization

For cognitively impaired users, effective interface design can be visualized as a design with fewer decisions, clearer visual cues and predictable navigation (Cha, 2024). Personalization driven by Artificial Intelligence (AI) extends this by allowing systems to tailor the complexity of exercises and to time an individual's own performance patterns (Liu et al., 2024). The trade-off is significant personal data collection, which raises legitimate privacy concerns. To combat this issue, transparency about data use should not be up for debate.

4. THE MASA FRAMEWORK

The foundations of MASA are built from the reviewed evidence. Its core components include a smart reminder system that delivers prompts at behaviorally optimal moments (Maggio et al., 2023), an adaptive focus timer grounded in attention regulation research (Skulmowski & Xu, 2021), gamified cognitive exercises aligned with CCT findings (Chan et al., 2024; Li et al., 2026) as well as an AI scheduling assistant that evolves simultaneously with the user's behavioral rhythms over time (Liu et al., 2024). The interface commits to minimalism by decreasing aesthetic complexity, meeting accessibility contrast standards and pursuing design processes that are iterative and user-centered. AI processing is distributed between on-device and cloud computation to balance responsiveness with privacy protection.

5. DISCUSSION AND CONCLUSION

In summary, this paper has provided substantial evidence that mobile cognitive interventions can meaningfully improve cognitive-related aspects such as working memory, sustained attention and executive function, particularly when individualized (Cha, 2024; Chan et al., 2024; Li et al., 2026). Instruments designed for people with ADHD prove significantly more

effective when partnered with techniques that modify actions, habits or routines (Gabarron et al., 2025).

However, limitations do remain. Most studies can be considered short-term and relatively small in terms of size and/or scope (Cha, 2024; Maggio et al., 2023). Furthermore, as stated in a systemic review by Wilson et al. (2022), gaps created by the understanding of the digital tools continue to exclude many older adults from accessing these tools. Not to mention, privacy concerns around data collection regarding behavior should be viewed in-depth and ample attention (Liu et al., 2024).

In this current state, the demanding needs are long-term trials, real-world outcome measures, and AI systems that genuinely earn user trust as well as offer explainability. MASA attempts to bridge credible science and everyday use through an effective implementation that tackles cognitive challenges.

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